



Informational Update – Vacuum Housing

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Summary

The vacuum housing upgrade is newly released. Since the release of the upgrade a few issues have been identified that should be shared. This information will be included in the next revision of the FSN “Vacuum Housing Retrofit Procedure” (GP12-204-FSN). This information is supplemental to the FSN and should be considered as an informational reference only.

The damping mounts can be damaged during unpacking of the hardware. This can be easily avoided with the simple procedure described below.

The plastic barbed fittings in housing manifold and in the pump heads can break during installation or removal of the vacuum tubing if not performed carefully. A procedure for replacing the fittings is described below.

During installation of the vacuum pumps into the housing it is critical that cables are connected correctly. Improper connection can result in time-consuming troubleshooting as well as damage to the vacuum pumps. The information below provides additional details and images that will be useful in ensuring proper cable connections. An additional recommended pump operation check along with troubleshooting tips is also described.

Removing the Pump Mount from the Inner Housing

The vacuum pump housing is shipped assembled. The vacuum pump mounting bracket and base assembly must be removed from the inner housing before assembling the pumps onto the bracket and base. The vibration damping mounts located on the bottom of the base are soft, which is good for reducing vibration and noise. Because they are soft, the metal inserts can be pulled out of the rubber (see Figure 1). This can occur during initial unpacking of the hardware or during subsequent servicing of the vacuum system that may require removal of the vacuum pumps.



Figure 1 - Insert Pulled Out of Damping Mount

The manufacturing process sometimes results in rubber “flash” on the surface of the lower insert (see Figure 2). The “flash” can cause the insert to adhere to the metal surface in the bottom of the inner housing. DO NOT pull on the base/bracket assembly if the damping mounts are adhered to the bottom of the inner housing as this could cause the inserts to pull out of the rubber mount. To avoid pulling the insert out of the damping mount, the insert should be carefully separated from the bottom of the inner housing if it is adhered. This can easily be accomplished during removal of the bracket/base assembly by loosening the four mounting screws without fully removing them and then pushing on the head of the screw with the driver (see Figure 3). This will apply force to the insert to separate it from the bottom of the inner housing. The bracket/base assembly should then be able to be easily removed without damaging the mounts.

If the damping mounts do become damaged, the mounts can be easily replaced. These are supplied as part of the Panther Fasteners Kit. The mounts can be removed and installed in the base of the assembly by hand. These should only be tightened snug by hand.

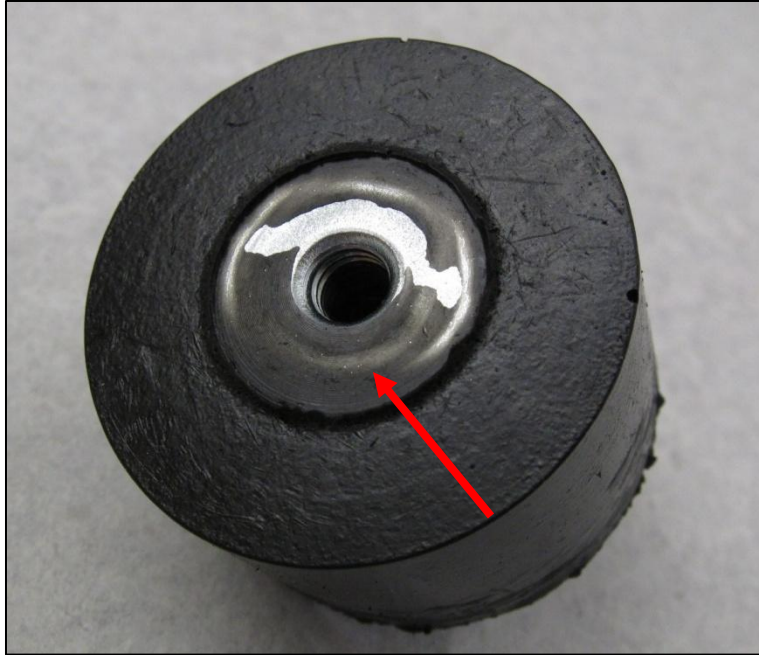


Figure 2 - Flash on Damping Mount Insert

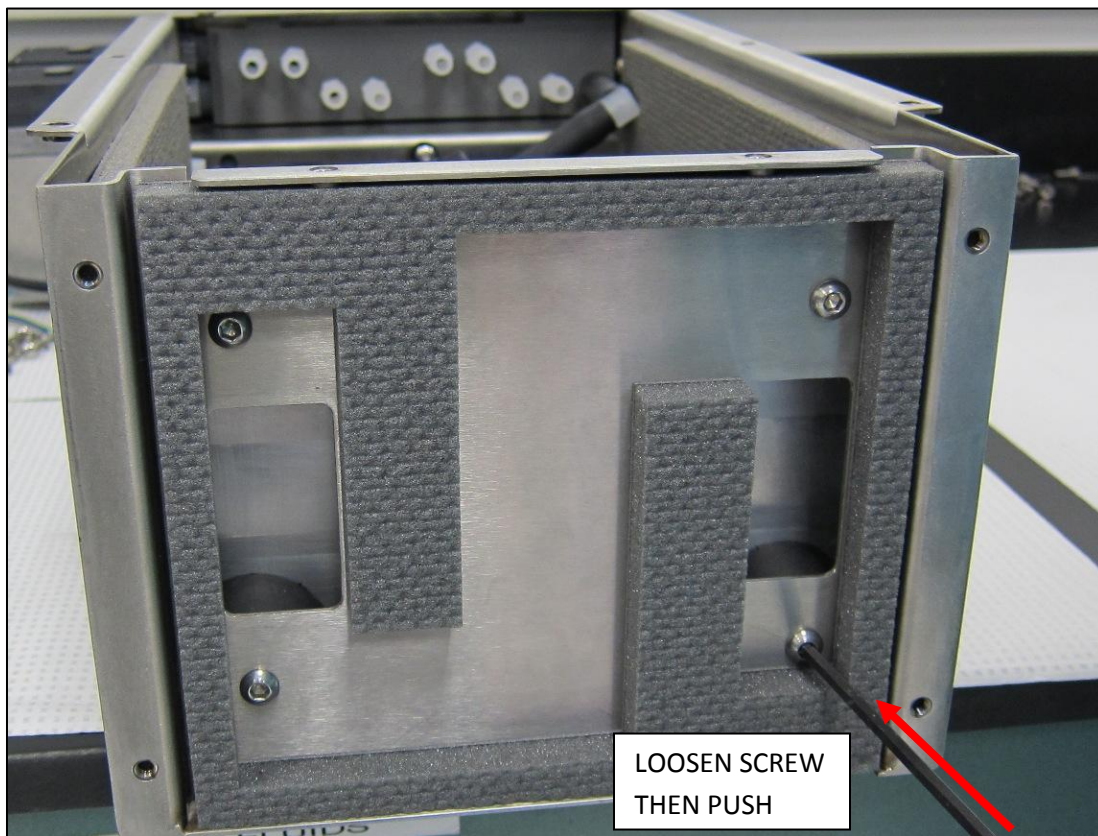


Figure 3 - Separating Damping Mounts from Bottom of Inner Housing

Broken Tube Fitting Replacement

Extra care should be taken when connecting and disconnecting the vacuum tubing to the barbed fittings in the housing manifold and pump heads. These fittings can break if too much bending force is applied. Stabilizing the fitting with one hand and applying twisting motion to the tubing will reduce the chance of damaging the fitting. If the fitting does break it can be removed with the extractor supplied in the Panther Fasteners Kit. Tap the extractor into the broken fitting with a hammer or mallet (see Figure 4). Then twist the extractor counterclockwise using an adjustable wrench to remove the fitting (see Figure 5). Install a new fitting from the Panther Fasteners Kit and tighten so that the fitting is oriented properly.

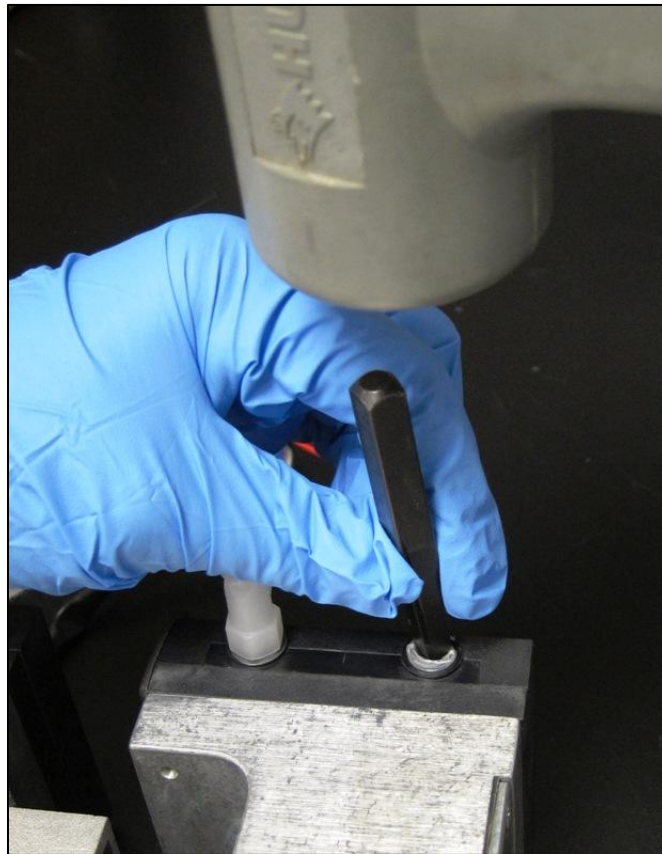


Figure 4 - Tap Extractor into Broken Fitting

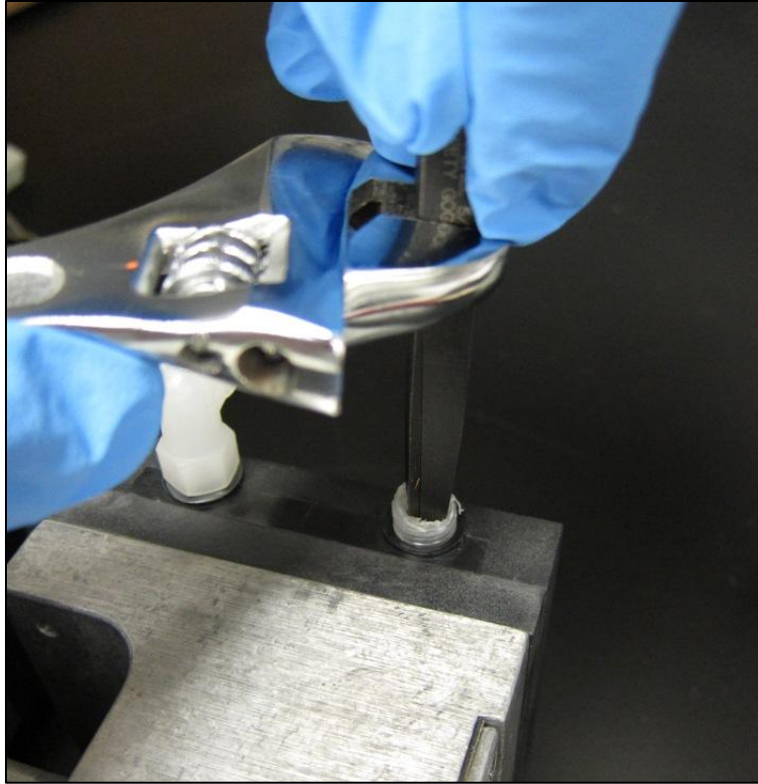


Figure 5 - Twist Extractor to Remove Broken Fitting

Proper Ribbon Cable Connection

It is critical that the ribbon cables that connect from the pump to the control board are connected correctly. The connectors on the control board have locking features that make it extremely difficult to remove the connectors if this becomes necessary. These locking features (two per connector) can be removed without removing the guide posts by using a small flat blade screwdriver or similar tool. The connector will still have adequate retention without the locking feature.

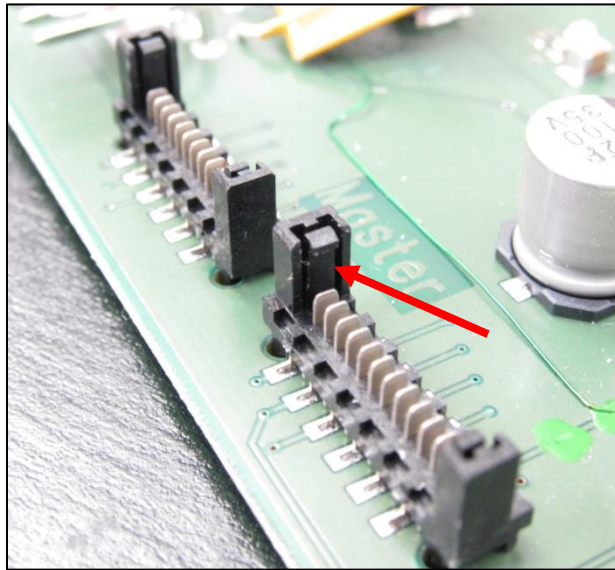


Figure 6 - Locking Feature on Control Board Connector

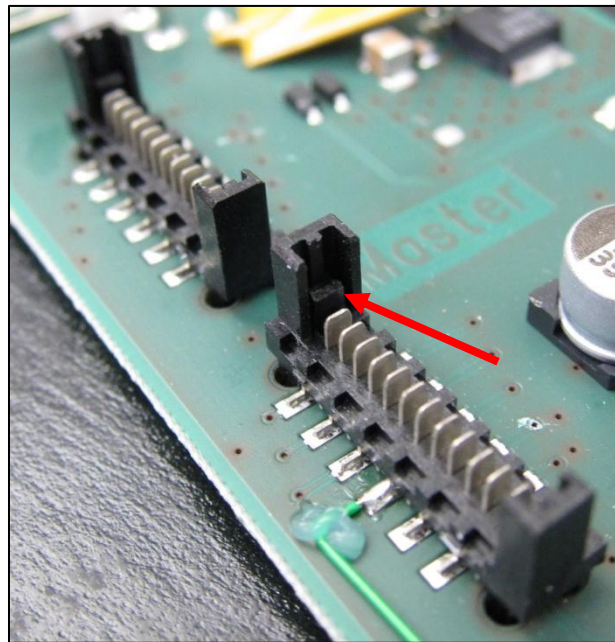


Figure 7 - Locking Feature Removed

The ribbon cables have the same connector on each end but the orientation is opposite relative to the direction the cables exit the connector. The side that connects to the vacuum pump controller (on the pump) will have the wide polarizing slot on the same side circuit one, identified with a stripe (see Figure 8).

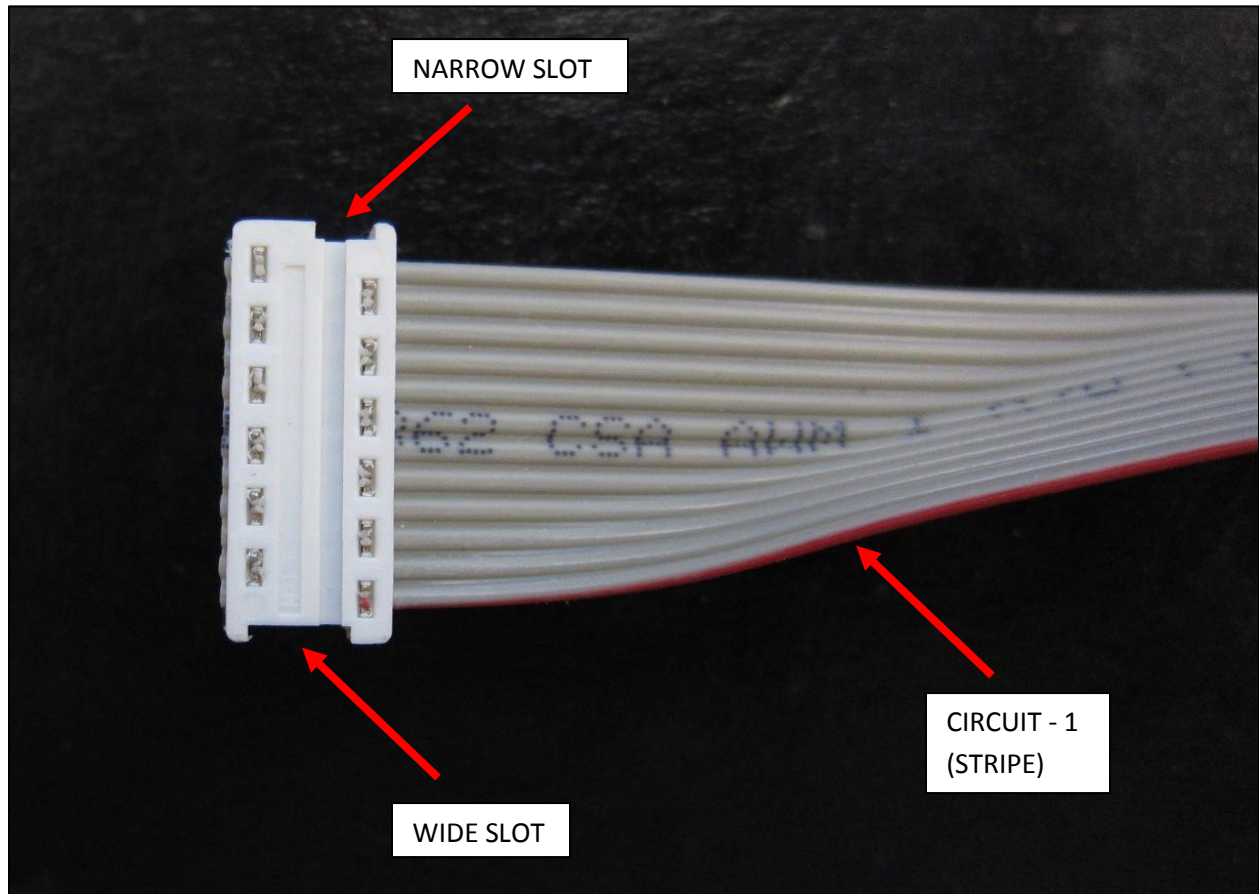


Figure 8 - Ribbon Cable Pump Connector

It is extremely important that the connection to the pump controller (on the pump) is correct so that the polarizing slots on the ribbon cable connector match up with the polarizing posts on the controller connector – wide slot with wide post, narrow slot with narrow post. Use care when making the connection so that the terminals are not damaged. When properly connected and to the correct end of the ribbon cable, the ribbon cable should exit the connector toward the pump (see Figure 9).

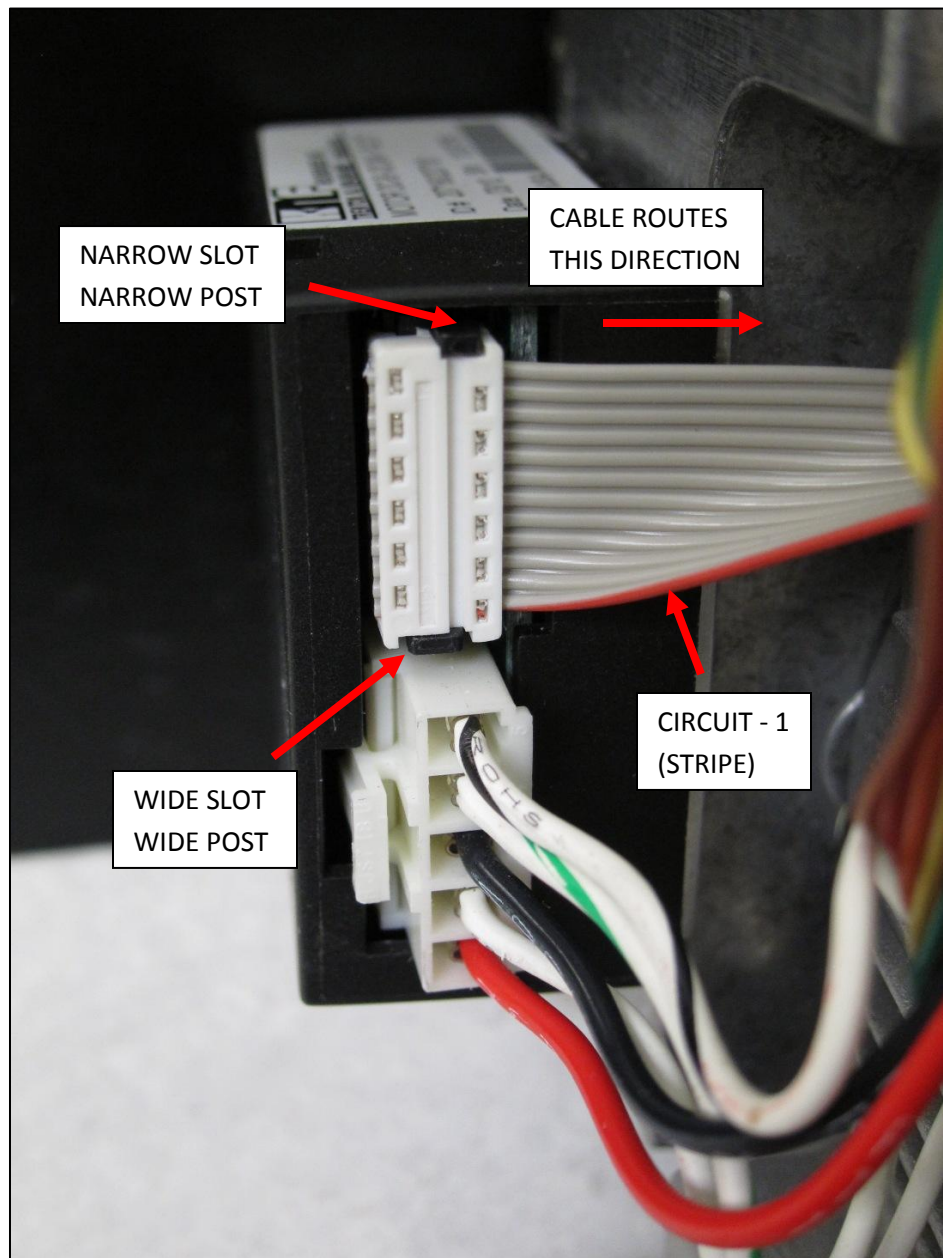


Figure 9 - Proper Ribbon Cable Connection to Pump Controller

The opposite end of the ribbon cable connects to the lower of the two connectors on the large control board (on the pump bracket) for each of the vacuum pumps. The ribbon cable must be flipped around so that the cable enters the connector from the right side and circuit 1, identified with the stripe, is orient up, which will be the same side as the wide polarizing slot and post on the connection (see Figure 10).

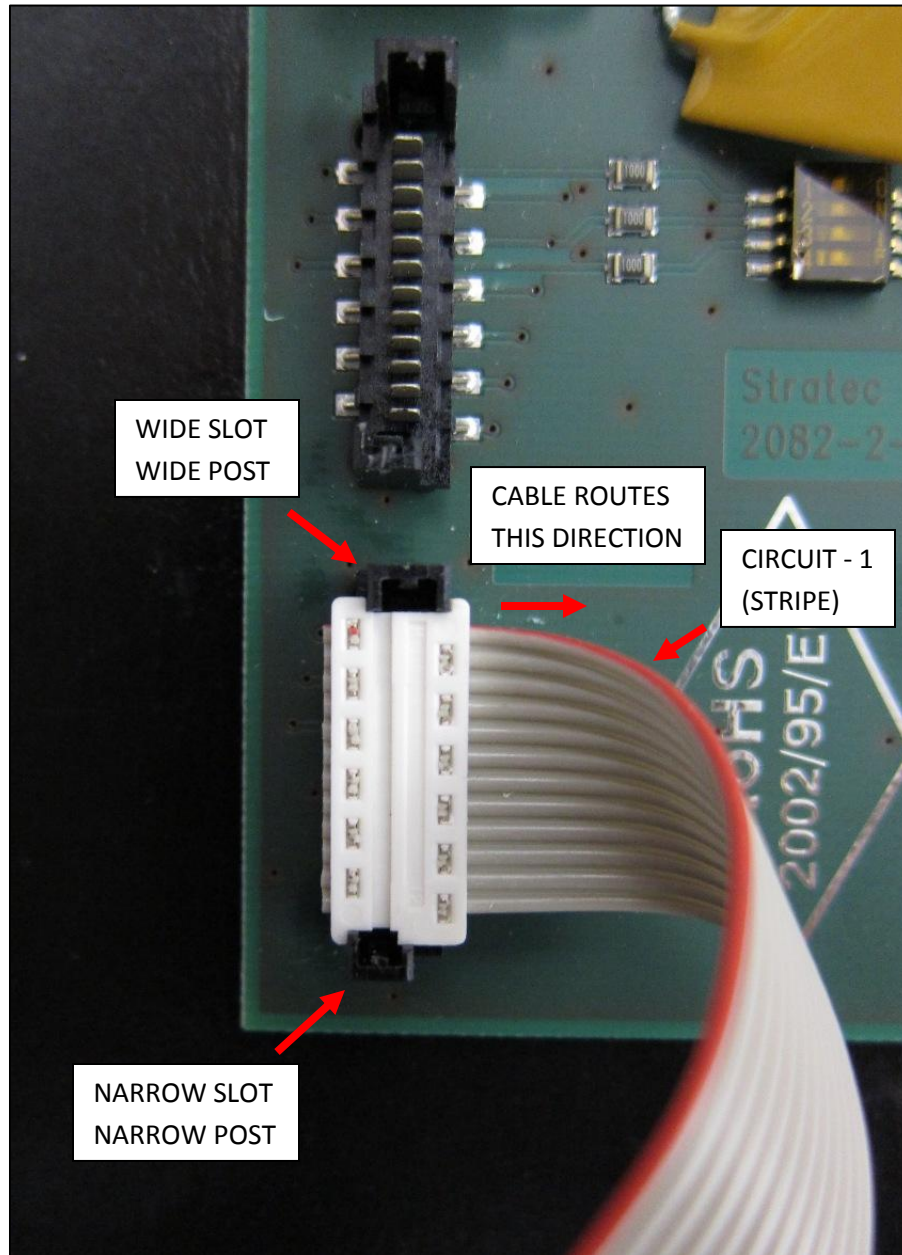


Figure 10 - Proper Connection of Ribbon Cable to Large Control Board

The multicolor ribbon cable from the pump connects to the upper connection on the on the large control board for each pump (see . The power cable from the pump connects to the uppermost connection on the large control board (see . The smaller signal connector from the pump is not connected at all.

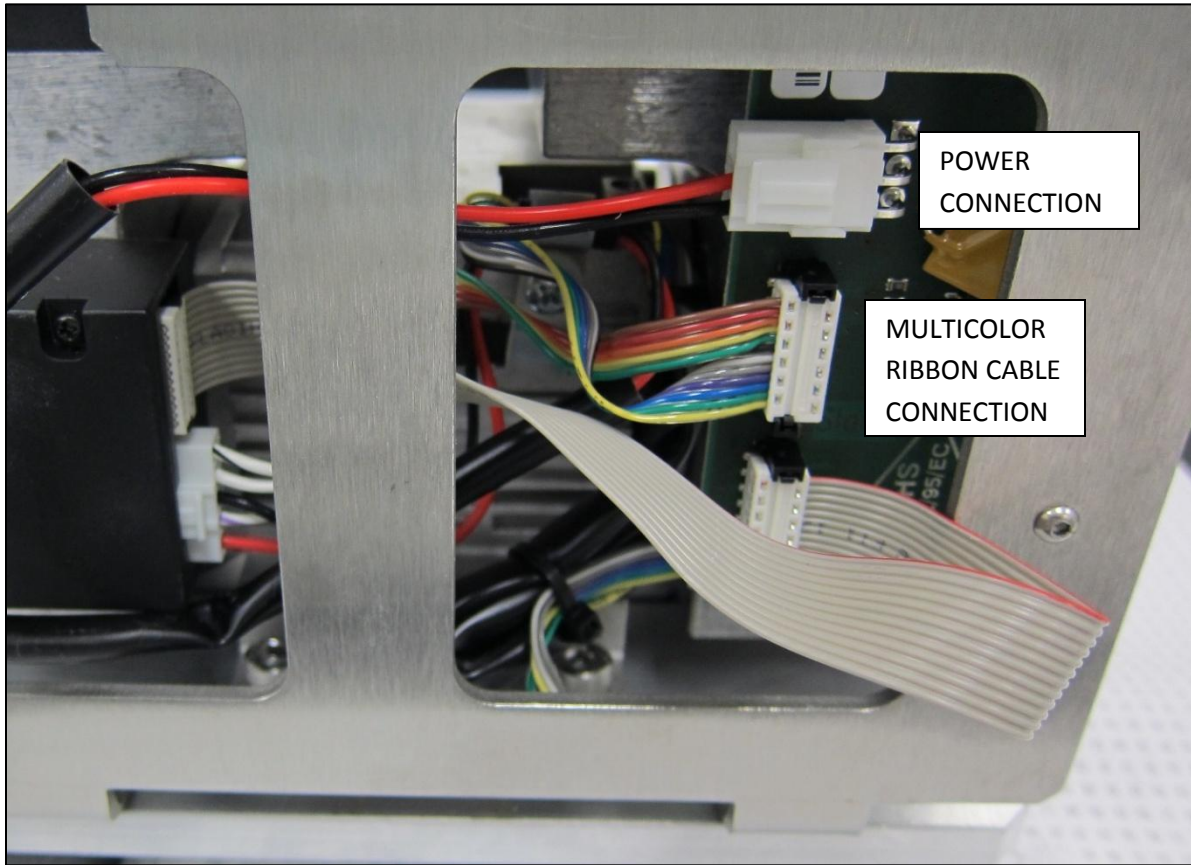


Figure 11 - Power and Ribbon Cable Connection

All cables including the unconnected smaller signal cable should be bundled and secured with a zip ties to prevent rubbing against any metal edges (see Figure 12). Also, the two square-shaped PTC fuses on the large control board may need to be bent over slightly to avoid interference with the pump bracket (see Figure 12).

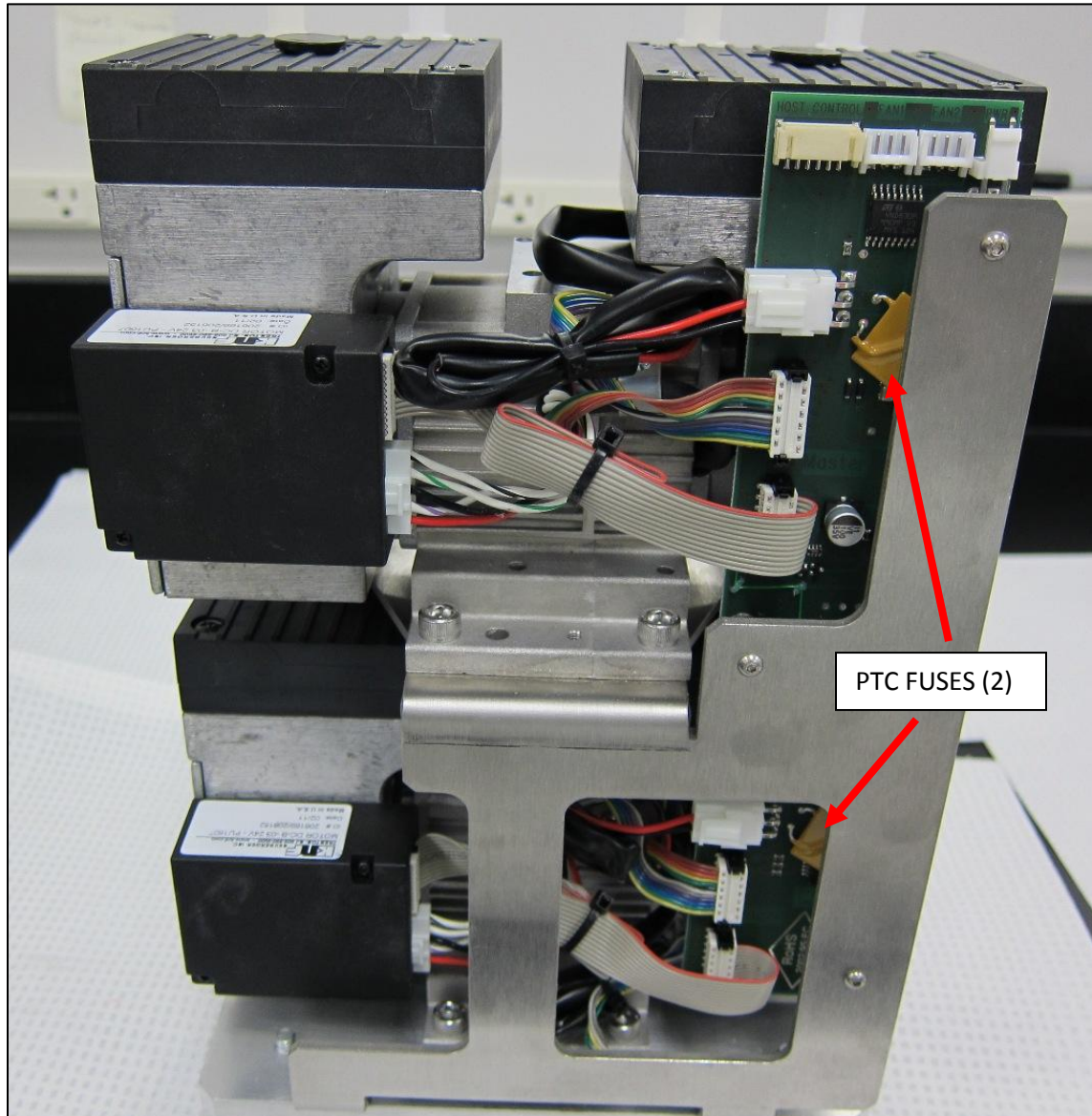


Figure 12 - Proper Cable Connection and Bundling and PTC Fuses

Pump Test and Troubleshooting

After the pumps have been mounted to the pump bracket/base assembly and after the control board had been mounted with all cables properly connected and secured (see Figure 12), it is **HIGHLY RECOMMENDED** to test the pumps to ensure that both pumps operate. This can be accomplished by connecting the pump control board to the instrument using the power and signal cables supplied with the vacuum housing kit. The new power supply cable (that has a connector) that connects to the instrument power supply must be installed (see GP12-204-FSN). The signal cable connects to the Cooling Module (see GP12-204-FSN). After connecting the power and signal cables, power on the Panther System and use Service Software to initialize and activate the vacuum pumps. Check to make sure both pumps are operational. Power off the Panther System after the check is complete and before disconnecting any cables.

NEVER CONNECT OR DISCONNECT ANY CABLES WHILE THE PANTHER SYSTEM IS POWERED ON.

If either pump does not operate check all cable connections to ensure proper connection. Check to ensure that none of the connector terminals are bent (see Figure 13).

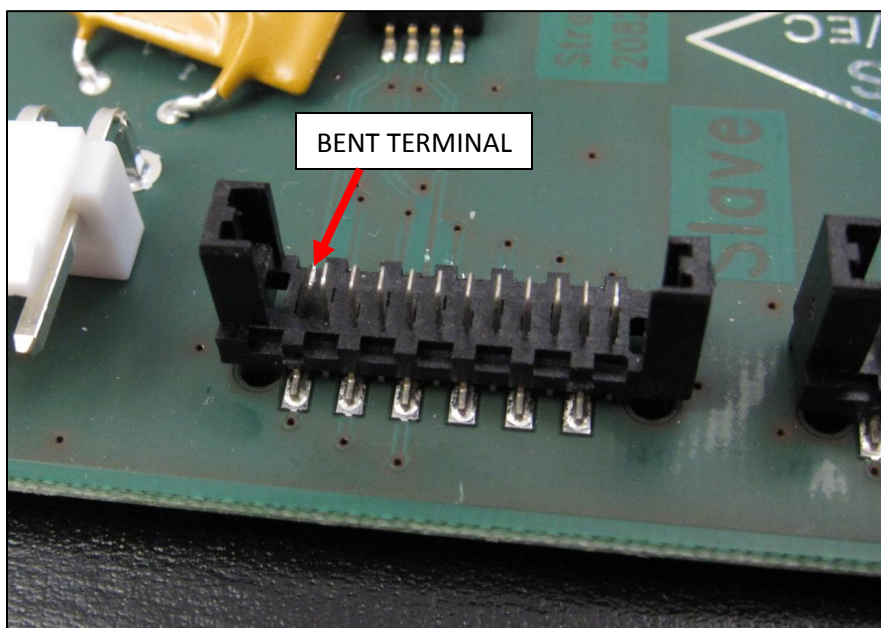


Figure 13 - Bent Connector Terminal

If one pump does not operate, it may be helpful to swap the top and bottom pumps. It is recommended to disconnect and reconnect all cables to ensure proper connections. If the same pump still does not operate, it may be a failed pump. Swapping ribbon cables can verify if either ribbon cable is faulty. If the non-operational pump is always the top pump or always the bottom pump then the large control board may be faulty.

After the vacuum housing installation and verification is complete, unlock the waste drawer using Panther Main Software and ensure that both pumps turn off.